.05 13.42 1.50 0.03 70.8750 134.2 0.4688 0.006614 66.1376
Residual 87.17 11.03 1.50 0.30 72.6417 110.3 0.4688 0.006453 64.5291
* Based on properties in the IMO NOx Monitoring Guidelines, resolution MEPC.103(49).
Table 2: Emissions calculations corresponding to 1.5 % fuel sulphur
CO2 SO2 Exh SO2/CO2 Exh S/C
% ppm4 ppm4/% m/m
Distillate 0.17% S 8 59.1 7.4 0.00197
Residual 2.70% S 8 939.7 117.5 0.03136
Distillate 1.5% S 8 528.5 66.1 0.01764
Residual 1.5% S 8 515.7 64.5 0.01721
Distillate 1.5% S 0.5 33.0 66.1 0.01764
Residual 1.5% S 0.5 32.2 64.5 0.01721
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SOURCE: MEPC.259(68).pdf
Residual 2.70% S 8 939.7 117.5 0.03136
Distillate 1.5% S 8 528.5 66.1 0.01764
Residual 1.5% S 8 515.7 64.5 0.01721
Distillate 1.5% S 0.5 33.0 66.1 0.01764
Residual 1.5% S 0.5 32.2 64.5 0.01721
SO2/CO2 ratio vs % sulphur in fuel
0
50
100
150
200
250
0,00 0,50 1,00 1,50 2,00 2,50 3,00 3,50 4,00 4,50 5,00
% m/m sulphur in fuel
SO2(ppm)/CO2(%)
Distillate fuel
Residual fuel
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4 Correspondence between 65 (ppm 4/%) SO2/CO2 and 6.0 g/kWh is demonstrated by showing that their S/C ratios are similar. This requires the additional assumption of a brake-specified fuel consumption value of 200 g/kWh. This is an appropriate average for marine diesel engines. The calculation is as follows:
S/Cfuel = brake-specific SO2
(MWS / MWSO2)
BSFC
(% carbon in fuel / 100)
brake-specific SO2 = 6.0 g/kW-hr
MWS = 32.065 g/mol
MWSO2 = 64.064 g/mol
BSFC = 200 g/kW- hr
% carbon in 1.5% sulphur fuel (from table 1) = 85.05% (distillate) or 87.17%
---CHUNK---
SOURCE: MEPC.259(68).pdf
(MWS / MWSO2)
BSFC
(% carbon in fuel / 100)
brake-specific SO2 = 6.0 g/kW-hr
MWS = 32.065 g/mol
MWSO2 = 64.064 g/mol
BSFC = 200 g/kW- hr
% carbon in 1.5% sulphur fuel (from table 1) = 85.05% (distillate) or 87.17%
(residual)
S/Cresidual fuel = 6.0
(32.065 / 64.064) = 0.01723 200
(87.17% / 100)
S/Cdistillate fuel = 6.0
(32.065 / 64.064) = 0.01765 200
(85.05% / 100)
Note 1: The S/C mass ratios calculated above, based on 6.0 g/kWh and 200 g/kWh BSFC, are both within 0.10% of the S/C mass ratios in the emissions table (Table 2). Therefore, 65 (ppm4/%) SO2/CO2 corresponds well to 6.0 g/kWh.
Note 2: The value of 6.0 g/kWh, hence the 200g/kWh brake-specified fuel consumption is taken from MARPOL Annex VI as adopted by the 1997 MARPOL Conference.
5 Thus, the working formulas are as follows:
For complete combustion = SO2 (ppm*) ■ 65 CO2 (%*)
For complete combustion = SO2 (ppm*) ■ 65 CO2 (%*) + (CO(ppm*)/10000) + (THC(ppm*)/10000)
* Note: gas concentrations must be sampled or converted to the same residual water content (e.g., fully wet, fully dry).
---CHUNK---

SOURCE: MEPC.259(68).pdf
For complete combustion = SO2 (ppm*) ■ 65 CO2 (%*) + (CO(ppm*)/10000) + (THC(ppm*)/10000)
* Note: gas concentrations must be sampled or converted to the same residual water content (e.g., fully wet, fully dry).
6 The following is the basis of using the (ppm 4/%) SO2/CO2 as the limit for determining compliance with regulation 14.1 or 14.4 of MARPOL Annex VI:
.1 This limit can be used to determine compliance from fuel oil burners that do not produce mechanical power.
.2 This limit can be used to determine compliance at any power output, including idle.
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.3 This limit only requires two gas concentration measurements at one sampling location.
.4 There is no need to measure any engine parameters such as engine speed, engine torque, engine